

Abstract

Title: Titan 72V Formula for Imperial Electric Racing Battery Pack Using Smart BMS

Summary: This project details the design, development, and optimization of a high-performance lithium-ion battery system engineered specifically for electric mobility and competitive electric racing applications. The primary objective of the work is to deliver a lightweight, efficient, and durable 72-volt power source capable of supporting sustained high-current discharge while maintaining rigorous safety standards and optimal thermal stability.

System Architecture & Technical Specifications:

- The battery pack utilizes a 20S-5P configuration constructed from high-discharge Lithium-ion 18650 cells to provide a robust nominal 72V output.
- The design emphasizes a modular structural architecture, enabling seamless scalability for various electric vehicle platforms ranging from e-bikes to high-speed electric karts.

Smart Management & Validation:

- The core of the system's safety and efficiency is the integration of a Smart Battery Management System (BMS) that features advanced cell balancing mechanisms.
- The Smart BMS facilitates the real-time monitoring of critical operational parameters, including State of Charge (SOC), individual cell voltages, and temperatures, utilizing CAN and Bluetooth communication protocols.
- Key performance indicators, such as SOC accuracy, charge/discharge efficiency, and thermal stability under extreme load conditions, were successfully validated through extensive MATLAB simulations and empirical lifecycle testing.

Impact:

- Ultimately, the resulting battery pack presents a highly resilient and efficient power source designed to meet the dynamic demands of Formula-style electric racing. By establishing new benchmarks in the power-to-weight ratio and overall system reliability, this project actively contributes to the continued evolution and advancement of sustainable electric propulsion technologies.